

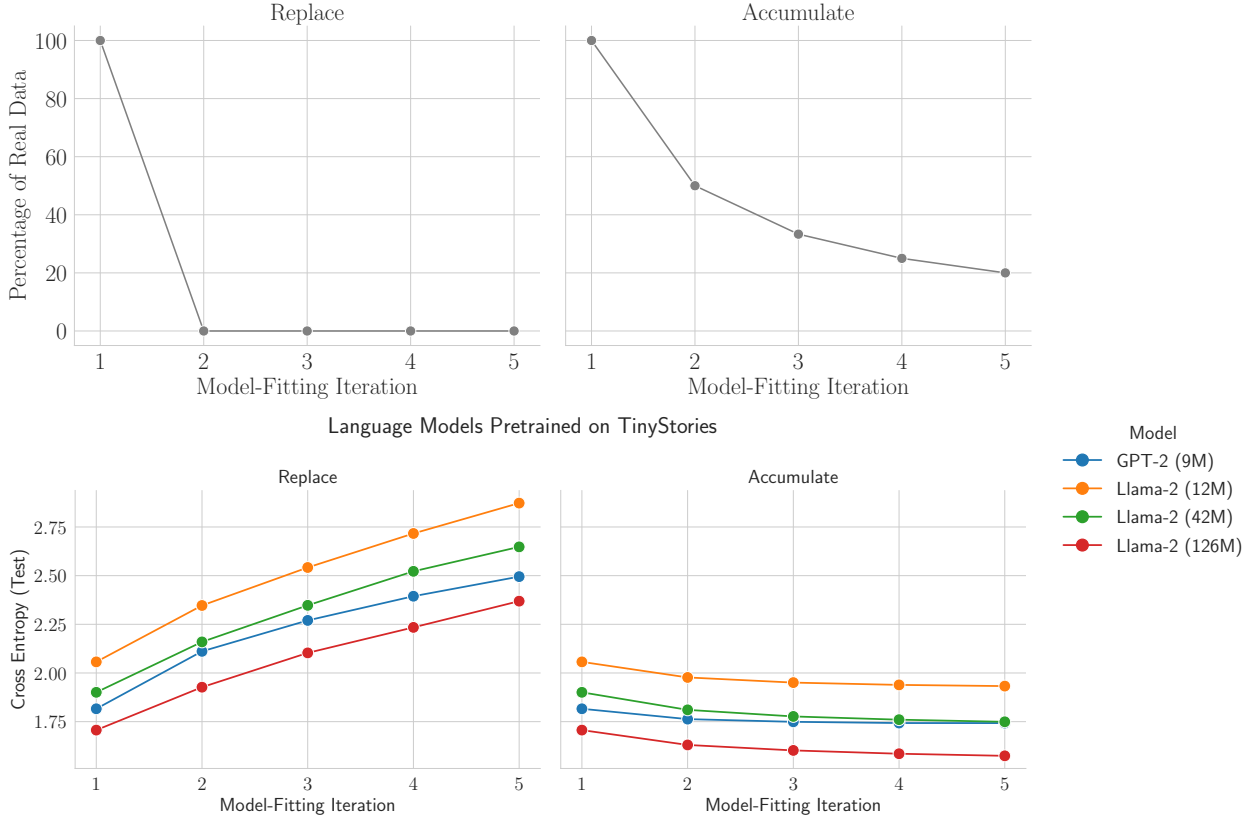


Figure 3. Dimensions of Consideration for Model-Data Feedback Loops: Propagation of Data Over Time and Proportion of Real Data Over Time. When data are *replaced* after each model-fitting iteration (left), the proportion of real data immediately becomes zero after the first iteration, whereas when data instead *accumulate* (right), the proportion of real data falls asymptotically to zero. Gerstgrasser et al. (2024); Kazdan et al. (2024b); Dey & Donoho (2024) showed that replacing data over time causes the population risk to diverge, whereas accumulating data avoids diverging population risk. In these works, synthetic data are assumed to grow linearly over time, contributing n samples per model-sampling iteration. Credit: The bottom figure is copied from Gerstgrasser et al. (2024) with permission.

fraction of training data which is synthetic does not vanish” and that “model collapse cannot generally be mitigated by simple adjustments [...] unless these strategies asymptotically remove all but a vanishing proportion of synthetic data from the training process.”

Results like these draw attention to what proportion of real data is necessary to avoid collapse, which is a valid consideration. However, correctly interpreting these results takes care. For instance, Bertrand et al. (2023)’s condition is *sufficient*, not necessary; it should *not* be understood as saying that model collapse is inevitable unless real data remains a non-zero proportion. Moreover, contrary work by Gerstgrasser et al. (2024), Marchi et al. (2024), and Kazdan et al. (2024b) established conceptually stronger guarantees: when data *accumulate*, but human data asymptotically occupy a vanishing fraction of the internet, the population risk will likely not diverge (in some cases, with an additional requirement that the rate of AI data generation does not grow super-linearly) (Fig. 3). Relatedly, Gillman et al. (2024)

also show that the proportion of real data can asymptotically approach zero using a function to “correct” synthetic data towards the real data distribution. Due to uncertainty about the rate of synthetic data generation, one potential solution is to sequester real training data for future use, when the internet still contains an abundance of human-generated data; however, frontier AI labs have already collected such data, meaning no additional work is required.

4.3. Model Training Assumptions

While multiple works claim that models do not collapse under certain settings or propose interventions to avoid collapse, we urge the explicit declaration of strong technical assumptions that are frequently unrealistic. As an example, Bertrand et al. (2023) and Gillman et al. (2024) study *iterative retraining*, where each model is initialized from its predecessor’s parameters and optimizer state. By making this assumption, each model is close to its predecessor; then, under an additional assumption that the first model is

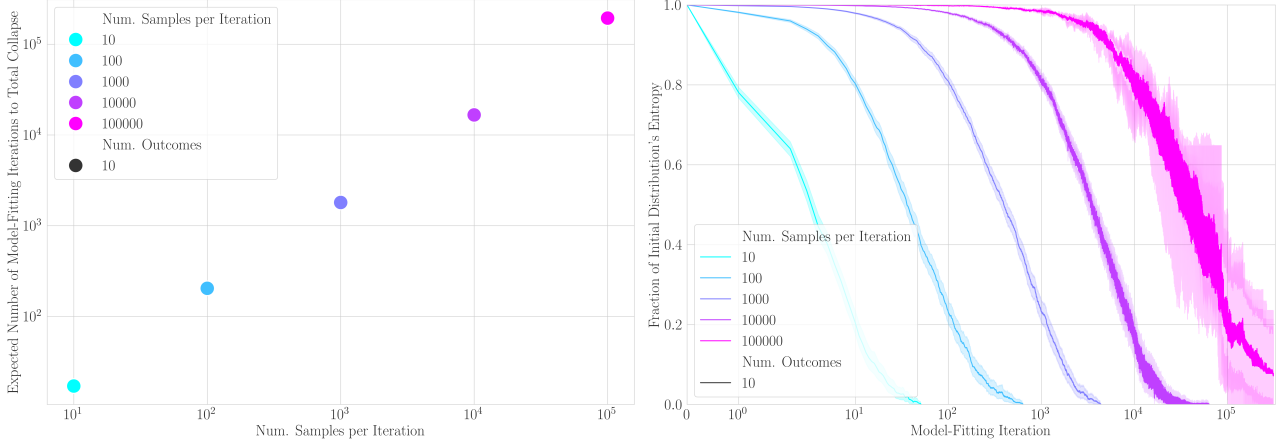


Figure 4. Dimension of Consideration for Model-Data Feedback Loops: Timescales of Collapse. Characterizing the timescale over which one should expect collapse is an underappreciated but crucial consideration. Focusing on the discrete model of Shumailov et al. (2023), the expected number of model-fitting iterations before total collapse is proportional to the number of data times the entropy of the initial data distribution at face value, this means that trillions of models can be trained before glimpsing the onset of collapse. However, total collapse is only the most extreme outcome; in this model, we additionally show how the entropy of the initial data distribution decays over time (right). Error bars are over 100 seeds (0 to 99, inclusive); for experimental details, see Sec. 4.4.

sufficiently high performing, since each subsequent model is close to its predecessor, model collapse can be avoided. However, to the best of our knowledge, iterative retraining has not been used for any frontier AI model, including OpenAI’s GPT-2 (Radford et al., 2019), GPT-3 (Brown et al., 2020), GPT-4 (Achiam et al., 2023), Anthropic’s Claude 1, 2 or 3, Google’s PaLM 1 (Chowdhery et al., 2022), PALM 2 (Anil et al., 2023) or Gemini (Team et al., 2024), DeepSeek’s V3 (DeepSeek-AI et al., 2024). Thus, Bertrand et al. (2023)’s sufficiency conditions for avoiding collapse are far from current practices.

For another example, Zhu et al. (2024) propose data editing as a collapse mitigation strategy and analyze a self-consuming linear model. Each model iteration fits $\hat{w}_n = X^\dagger \tilde{Y}_n$, where X are fixed Gaussian covariates, and generates synthetic data $\hat{Y}_{n+1} = X\hat{w}_n + E_{n+1}$ where E_{n+1} are Gaussian errors and $\tilde{Y}_n$ are regression targets sampled from the previous generation’s training targets with edits from the prior generations synthetic data:

$$\tilde{Y}_n^\top = M_{n-1} \hat{Y}_n + (1 - M_{n-1}) \tilde{Y}_{n-1}$$

Here, M_k is a diagonal matrix of 1’s or 0’s indicating whether to replace or not replace a label with a synthetic value. While Zhu et al. (2024) claim that this prevents model collapse, the proof of their test error bound (Theorem 2) assumes that $\|M_i\| = \eta \|M_{i-1}\|$ for some constant $\eta \in (0, 1)$, meaning that the number of edits decreases by at least some fixed proportion each generation. This geometric decay guarantees that the total number of edits at any given generation is finite, and their proof fails without this. However, training GPT-2 on the Natural-Instructions dataset (Mishra

et al., 2021) yields only a slight decline in edit percentage (Zhu et al., 2024). Moreover, if the number of edits is finite, each generation trains on mostly real data.

4.4. Timescales of Collapse

Another key dimension of consideration is the timescale of model deterioration, which is often omitted. For example, Shumailov et al. (2023) introduced a simple theoretical setting for studying model collapse: a discrete distribution (picture a histogram) with N outcomes (atoms):

$$p^{(0)}(x) = \sum_{n=1}^N w_n \delta_{x_n}(x),$$

where $\sum_n w_n = 1$. If one sequentially draws D data from this distribution and computes a new distribution based on the empirical proportions, then this process forms a Markov chain with N absorbing states, each corresponding to a totally collapsed distribution, i.e., a distribution comprised of exactly one outcome. Consequently, one can use standard results from absorbing Markov chains to show that this simple process *must* collapse.

However, one can go beyond a guarantee of collapse and ask: *how many model-fitting iterations can we survive before total collapse consumes us?* Again using standard results (Brydges, 2009), the expected number of model-fitting iterations before total collapse is:

$$\mathbb{E}[\text{Model Iterations Till Total Collapse}] \propto D H[p^{(0)}],$$

where $H[\cdot]$ is the Shannon entropy. For intuition, if our starting distribution is uniform, the entropy is $\log(N)$ and

thus the expected number of model-fitting iterations before total collapse is $D \log(N)$. We confirmed this claim using numerical simulations starting from uniform distributions (Fig. 4 Left). We also numerically simulated how quickly the entropy of $\hat{p}^{(t)}$ falls relative to the entropy of $p^{(0)}(x)$ to provide a description of the process more nuanced than just total collapse (Fig. 4 Right), since one might be interested in how quickly tail information is lost, not just how quickly total collapse arrives.

While this mathematical model is simple, for the sake of argument, if we take this model at face value, we realize total model collapse poses virtually no present threat. This is because the number of real public text data alone is on the order a quadrillion tokens (Villalobos et al., 2024), and text, image, video and agentic data are high dimensional with large entropy, meaning total collapse occurs so imperceptibly slowly that humanity could train trillions of models before noticing the onset of collapse. However, this highlights a more general point: characterizing the timescale over which one should expect model collapse to occur is an underappreciated but crucial consideration for describing the model collapse profile.

Several authors do characterize timescales. Suresh et al. (2024) give an exact formulation of model collapse rate in the fundamental setting of recursive maximum likelihood estimation for discrete distributions and Gaussian mixtures; they find that for $\text{Bern}(\mu)$, $\text{Pois}(\lambda)$, and Gaussian mixtures with shared variance σ^2 distributions, the parameters μ , λ , and σ collapse to 0 exponentially in the number of generations. While Seddik et al. (2024) previously controlled the total collapse probability in both the fully synthetic and partially-synthetic recursive training regimes, Suresh et al. (2024) further control the number of unique symbols after k generations in the fully synthetic regime. Lastly, Kazdan et al. (2024b) note, in the context of kernel density estimators with fixed bandwidths, that the negative log likelihood does diverge asymptotically, although “this occurs at a rate so glacial that it doesn’t pose a practical concern.”

5. Is Model Collapse a Threat?

In conclusion, is model collapse a threat? In our view, model collapse is a multifaceted phenomenon, and a single answer is not possible. By taking a realism-weighted average of different papers’ results, we synthesize our own forecast of model collapse under the different definitions:

Population risk will not increase catastrophically or diverge asymptotically. Given the increase in pretraining dataset size and quality, we argue that models training on accumulating synthetic data alongside real data will not suffer from catastrophic population risk increase or diverging population risk. The jury is still out on whether the proportion

of real data relative to available data will approach zero.

Real tail data and modes will be lost, but how many and how quickly is unclear. Loss of diversity is a real issue, with disproportionate harms oftentimes born by subgroups. It is unclear how much of the tail we will lose or which of the real data modes will become entangled, and how synthetic data affect such changes that already occur naturally. We strongly encourage more research regarding coverage and mode collapse prevention strategies for realistic settings, building on prior work such as Hashimoto et al. (2018); Ensign et al. (2018); Taori & Hashimoto (2023).

Scaling laws may change with the introduction of synthetic data. The precise nature of these changes under realistic conditions remains to be determined. Synthetic data could potentially remove what some researchers describe as a data bottleneck, but this benefit might come at the cost of altered scaling law parameters. We encourage further research into how synthetic data affects scaling behaviors to better characterize likely future outcomes.

We emphasize that while real threats do exist, the popular perception that synthetic data on the internet will render future frontier AI models pretrained on web-scale data useless is likely unrealistic since such failures appear in conditions that do not faithfully match what is actually done in practice. Subtle degradations in data distributions might still insidiously occur, such as loss of real tail data, and future work should aim to explore what can be used to counter such outcomes.

6. Alternative Views

One may feel the conditions we identify in Section 3 are inaccurate, disproportionately emphasized, likely to change, or ignorant of important settings other than pre-training. One might also believe that the identified model collapse definitions in Section 2 do not fully encompass the literature or are inaccurately applied in Figure 2. Finally, a concerned bystander could argue that the benefits of being over-cautious outweigh the costs: if society fails to anticipate model collapse by allowing wanton generation of poor-quality synthetic data, then the internet could become flooded with low-quality samples that preclude future progress. These are valid points, and we look forward to engaging with researchers and policymakers to better identify what matters to them and what the future looks like in those directions.